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Started	Wednesday, 28 October 2026, 9:13 PM
Completed	Wednesday, 28 October 2026, 9:15 PM
Duration	2 mins 13 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

a. It is less efficient than custom sorting algorithms
b. It sorts data out of the box efficiently
c. It only works with integer arrays
d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently

Question **2**

Complete

Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 80,150,200,470,40,90,300,400,70,120
b. 200,470,80,150,40,90,300,400,70,120
c. 40,70,80,90,120,150,200,300,400,470
d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **3**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By comparing elements sequentially
b. By sorting data in a single pass
c. By using the bubble-up method
d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **4**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Sorting
c. Merging
d. Searching

The correct answer is:
Complexity

Question **5**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It is based on the divide-and-conquer approach
b. It does not require recursion
c. It is less efficient than bubble sort
d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

a. Merge Sort
b. Heap Sort
c. Bubble Sort
d. Quick Sort

The correct answer is: Bubble Sort

Question **7**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

a. Quick sort

b. Heap sort
c. Bubble sort
d. Insertion sort

The correct answer is:
Insertion sort

Question **8**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of duplicate keys
- b. A sorting algorithm is stable if it preserves the order of all keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data in a single pass
- b. Rearranging data without sorting
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort cannot handle duplicate elements
- b. Bubble Sort does not guarantee sorted order
- c. Bubble Sort is less efficient for large lists
- d. Bubble Sort is difficult to implement

The correct answer is: Bubble Sort is less efficient for large lists

Question **12**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

- a. Sorting helps in finding duplicates quickly
- b. Sorting makes it harder to search for items
- c. Sorting is rarely used in programming
- d. Sorting decreases the efficiency of selection operations

The correct answer is: Sorting helps in finding duplicates quickly

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Binary Search
- c. Linear Search
- d. Merge Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Comparing each element with the others
- c. Dividing the input into smaller subproblems
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

- a. Each part is sorted independently
- b. The parts are ignored
- c. The parts are merged without sorting
- d. Each part is searched for a specific element

The correct answer is: Each part is sorted independently